

Homework 2: 23i-0612 Rashim Ibrar

$$A \in \mathbb{R}^{2 \times 2}$$

we can consider A as $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

For eigenvalues,

$$A^T = \lambda A$$

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \lambda \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$a = \lambda a$$

$$c = \lambda b$$

$$b = \lambda c$$

$$d = \lambda d$$

$$\cancel{\lambda}$$

DALMATIAN

Date:

$$c = \lambda b \text{ and } b = \lambda c$$

$$b = \lambda^2 b$$

$$\lambda^2 = 1$$

$$\lambda = +1 \text{ or } \lambda = -1$$

For eigenvectors:

For $\lambda = +1$

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A = \begin{pmatrix} a & b \\ b & d \end{pmatrix} \text{ as } b = c$$

this is symmetric

For $\lambda = -1$

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} = - \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$a = -a \quad c = -b \quad d = -d \quad \text{but } a \neq 0 \text{ and } d \neq 0$$

So

$$A = \begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix}$$

Skew matrix

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Thus, the matrix is diagonalizable because there are two distinct eigenvalues $\lambda=1$ and $\lambda=-1$ with a corresponding eigenvector

The eigenbasis is:

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}$$

Problem 2:

$$D(f) = f'$$

$$f' = f \lambda$$

$$f'(u) = f(u) \lambda$$

$$\lambda = \frac{f'(u)}{f(u)}$$

Integrate both sides

$$\int \lambda = \int \frac{f'(u)}{f(u)}$$

$$\int \lambda = \int \frac{1}{f(u)} \times f'(u)$$

$$\ln + c = \ln |f(u)|$$
$$e^{\ln + c} = f(u)$$

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$$f(x) = e^c \cdot e^{dx}$$

$A = e^c$ as constant

$$f(x) = ~~Ae^c~~ A e^{dx}$$

d can be any real number
thus all real numbers are eigenvalues
& $D(f) = f'$

Problem 3

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

for eigenvalues:

$$|A - I\lambda| = 0$$

$$\left| \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$$

$$[(1-\lambda)(3-\lambda)] - 8 = 0$$

$$3 - \lambda - 3\lambda + \lambda^2 - 8 = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$\lambda_2 = 5 \quad \text{or} \quad \lambda_1 = -1$$

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For Eigenvectors:

$$\lambda_2 = 5$$
$$\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$$

$$\lambda_1 = -1$$

$$\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

$$D = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \text{Diagonal matrix}$$

$$P = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$$

Problem 4

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

For eigenvalues

Already found in previous hw.

$$\lambda_2 = 1 \quad (2 \text{ multiplicity})$$
$$\lambda_1 = 0$$

For eigenvectors

$$\lambda_2 = 1$$
$$\text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$$

$$\lambda_1 = 0$$
$$\text{span} \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$$

Date:

This matrix is not diagonalizable as it is a 3×3 matrix with only two linearly independent columns, we need all 3 to be L.I.

Problem 5:

$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

For eigenvalues:

$$|A - \lambda I| = 0$$

$$\left| \begin{pmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right|$$

$$\left| \begin{array}{ccc|c} 2-\lambda & 4 & 3 & \\ -4 & -6-\lambda & -3 & \\ 3 & 3 & 1-\lambda & \end{array} \right| = 0$$

~~$$(2-\lambda) [(-6-\lambda)(1-\lambda) - (-3)(3)] - 4 [(2-\lambda)(1-\lambda)]$$~~

~~$$+ 3 [(-4)(3) - (-6-\lambda)(3)]$$~~

$$(2-\lambda) [(-6-\lambda)(1-\lambda) - (-3)(3)] - 4 [-4(1-\lambda) - (-3)(3)]$$

$$+ 3 [(-4)(3) - (-6-\lambda)(3)]$$

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$$(2-\lambda) \begin{bmatrix} -6+6\lambda & -\lambda+d^2+9 \end{bmatrix} - 4 \begin{bmatrix} -4+4\lambda-9 \end{bmatrix} + 3 \begin{bmatrix} -12-(-18-3\lambda) \end{bmatrix}$$
$$(2-\lambda) \begin{bmatrix} d^2+5d+6 \end{bmatrix} - 4 \begin{bmatrix} 4d+5 \end{bmatrix} + 3 \begin{bmatrix} -12+18+3\lambda \end{bmatrix}$$

$$\cancel{2\lambda^2+10\lambda+6-\lambda^3-5\lambda^2-3\lambda-16\lambda-20+18+9\lambda=0}$$
$$\cancel{= -\lambda^3-3\lambda^2+9\lambda-26=0}$$

$$2\lambda^2+10\lambda+6-\lambda^3-5\lambda^2-3\lambda-16\lambda-20+18+9\lambda$$
$$= -\lambda^3-3\lambda^2+9\lambda-4=0$$

$$\cancel{-\lambda^3-3\lambda^2+4}$$
$$\lambda^3+3\lambda^2-4=0$$

$$\lambda_2 = 1, \lambda_1 = -2$$

For eigenvectors:

$$\text{For } \lambda_2 = 1$$

$$(A - I\lambda)u = 0$$

$$\begin{bmatrix} 2-\lambda & 4 & 3 \\ -4 & -6-\lambda & -3 \\ 8 & 3 & 1-\lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_2 = 1$$

$$\begin{bmatrix} 1 & 4 & 3 \\ -4 & -7 & -3 \\ 8 & 3 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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$$2 \begin{bmatrix} 1 & 4 & 3 \\ 0 & 9 & -9 \\ 3 & 3 & 0 \end{bmatrix} \text{ By } R_2 + 4R_1$$

$$2 \begin{bmatrix} 1 & 4 & 3 \\ 0 & 9 & 9 \\ 0 & -9 & -9 \end{bmatrix} \text{ By } R_2 - 3R_1$$

$$2 \begin{bmatrix} 1 & 4 & 3 \\ 0 & 9 & 9 \\ 0 & 0 & 0 \end{bmatrix} \text{ By } R_3 + R_2$$

$$2 \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ By } \frac{1}{9} \times R_2$$

$$2 \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ By } R_1 - 4R_2$$

$$u_1 - u_3 = 0$$

$$u_2 + u_3 = 0$$

$$u_1 = u_3$$

$$u_2 = -u_3$$

u_3 is

free

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} u_3$$

Span

$$= \left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

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$$R_1 \Delta_2 = -2$$

$$\begin{bmatrix} 2 & -1 & 4 & 3 & | & 0 \\ -4 & -6 & -1 & -3 & | & 0 \\ 3 & 3 & 1 & -1 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 & 3 & | & 0 \\ -4 & -4 & -3 & | & 0 \\ 3 & 3 & 3 & | & 0 \end{bmatrix}$$

$$2 \begin{bmatrix} 4 & 4 & 3 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 3 & 3 & 3 & | & 0 \end{bmatrix} \text{ By } R_2 + R_1$$

$$2 \begin{bmatrix} 1 & 1 & 3/4 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 3 & 3 & 3 & | & 0 \end{bmatrix} \text{ By } R_1 \times \frac{1}{4}$$

$$2 \begin{bmatrix} 1 & 1 & 3/4 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 3/4 & | & 0 \end{bmatrix} \text{ By } R_3 - 3R_1$$

$$2 \begin{bmatrix} 1 & 1 & 3/4 & | & 0 \\ 0 & 0 & 3/4 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \text{ Swap } R_3 \text{ and } R_2$$

$$2 \begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \begin{array}{l} \sim \text{By } R_1 - R_2 \\ \sim \text{By } \frac{1}{3} \times R_2 \end{array}$$

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$$u_1 + u_2 = 0$$

$$u_3 = 0$$

$$u_1 = -u_2$$

~~u_2 is free~~ $\mathbb{R}$

consider u_1 as free

$$\begin{pmatrix} u_1 \\ -u_1 \\ 0 \end{pmatrix}$$

$$\text{Span} \left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}$$

we only have two vectors, we need 3.
 $\rightarrow$ has a multiplicity of 2 so we can find another vector associated with it. Thus, not diagonalizable.

$$\begin{bmatrix} 1 & 1 & 3/4 \\ 0 & 0 & 1 \end{bmatrix}$$